

Engineering Practices in Mathematics and Computing

- 1.1 Course Number: MA122
- 1.2 Contact Hours: 16 hours Credits: 5 (L-T-P: 1-0-2)
- 1.3 Semester-offered: (AY 2024-25) Even
- 1.4 Prerequisite: None
- 1.5 Course Instructor: Dr. Sudeep Kundu, Dr. Manoj K. Rajpoot

2. Objective: This course provides an introduction to fundamental computing principles and programming concepts. Students use the high-level programming language, MATLAB to develop and implement programs to solve engineering problems. Basic programming concepts covered include algorithm design, data types, flow control, functions, sorting, plotting, simulation, and numerical methods

3. Course Content:

Unit-wise distribution of content and number of lectures

Unit	Topics	Sub-topic	Lectures
1	Introduction to MATLAB	The MATLAB Environment, MATLAB Basics: Variables, Numbers, Operators, Expressions, Input and output, Vectors, Arrays/Matrices.	4
2	Data Processing and Plotting	Files and File Management: Import/Export, Basic 2D, 3D plots, Graphic handling, Interpolation.	4
3	Programming with MATLAB	Conditional Statements, Loops, MATLAB Programs: Programming and Debugging, Applications of MATLAB Programming.	4
4	Computing with MATLAB	Solving system of algebraic equations, Basic Symbolic Calculus, Numerical Integration, Solutions to Differential Equations, and Case studies of basic engineering problems.	4
		Total	16

4. Readings

4.1 Textbook: *Engineering computation with MATLAB*, 2nd Ed., D. M. Smith, Pearson Education, Inc., (2010).

4.2 Reference books: *MATLAB® for Engineers*, 3rd Ed., Holly Moore, Pearson Education, Inc., (2012).

5. Outcome of the Course:

- i. To get familiar with Matlab programming tools.
- ii. To apply mathematical skills to solve physical and engineering problems using Matlab.